Bio-O-Ton Master table – Read me

Revised short version | 13 October 2026 | Recording-time policy v4

Path to Master Table: **/lsdf01/lsdf/kit/ipf/projects/Bio-O-Ton/Bio_O_Ton_Master.csv**

Key columns

Column	Meaning
dawn_chorus_id	Recording ID.
lon, lat	WGS84 coordinates in decimal degrees.
grid_100m_id, grid_10m_id	Assigned grid cells. A grid ID only identifies the spatial cell and does not imply that LRT formation data are available for that cell.
datetime_local	German local recording date and clock time in Europe/Berlin, without timezone suffix. Exact format: %Y-%m-%d %H:%M:%S.
inside_lrt_polygon	True if the recording coordinate itself lies within an included LRT polygon. Different from the majority formation of a grid cell.
grid_100m_has_majority_formation, grid_10m_has_majority_formation	True if a majority LRT formation is assigned to the relevant grid cell. Does not imply the recording point lies within that polygon.

Finding clean data

Suggested filters	Applies to the relevant analysis domain
time_has_issues=False	Validated recording clock. Source provenance alone is not an issue.
sound_exists=True; duration_seconds >= 40; sound_has_issues=False	Usable audio. The 40 s threshold may be adjusted.
weather_point_exists=True; weather_point_has_issues=False	Weather file available and checked.
sentinel_exists=True; sentinel_has_issues=False	Sentinel-2 file available and checked.

These are domain-specific filters, not a requirement to use all domains. Inventory and QC can change as data are repaired or extended.

Finding data for training

Task	Suggested filter
Sound in grid cells with a majority LRT formation	grid_100m_has_majority_formation=True
Sound with coordinates directly inside LRT polygons	inside_lrt_polygon=True

Relevant secondary columns

Column	Meaning
_exists (sound, weather, sentinel)	True means a corresponding file was found in the LSDF inventory; not a live check.
_has_issues (sound, weather, sentinel, time)	True for a real problem, missing required data or a missing validation result; flags apply to their own domain.
_issue_codes	Machine-readable cause of a domain issue; see error codes below.
timestamp_source	localtimes: explicitly supplied recording clock; datetime: source fallback used when localtimes is missing. In the inspected German Dawn Chorus export, the Z/+00:00 suffix of these fallback values is misleading: its clock digits are German local time, not UTC. Both sources retain their labels. Valid corrected datetime fallbacks have time_has_issues=False.
duration_seconds	Checked audio duration in seconds; expected 59–61 s. The recommended clean filter is at least 40 s; empty (0 s) files are excluded.

Recording-time verification (13 October 2026)

The 3,288 fallback records were reconciled against the original CSV and converted consistently in Europe/Berlin with date-dependent CET/CEST, including DST checks. Four local dates and one UTC calendar date changed compared with the old master. Four weather products and one Sentinel product require renewed date-specific checks. 14 records were independently spot-checked against the Dawn Chorus website; this is not independent website verification of every record.

The previously documented “UTC-based fallback may be up to two hours wrong” statement no longer describes the zero-offset fallback class investigated here. `timestamp_source` remains provenance; `time_has_issues` is based on timestamp validity and QC. Existing independent `localtimes`/reference conflicts remain flagged.

Error codes

Domain / code	Meaning / suggested use
Sound: missing_file	Audio file absent; unavailable for analysis.
empty_file, no_audio_stream_found, no_audio_frames_decoded	No usable audio found; exclude pending repair or replacement.
pyav_decode_failed, invalid_or_missing_sample_rate, invalid_or_missing_channel_count, invalid_duration, duration_unavailable	Audio properties could not be reliably checked; review.
multiple_audio_streams	More than one audio stream; select and verify the intended stream.
filename_does_not_match_<id>_audio_pattern	File-to-recording-ID association needs clarification.
source_stat_failed, copy_failed, WinError	File access or copy error; inspect before assuming corruption.
Point weather: missing_file	Point-weather file absent.
missing_value	At least one weather value missing; informational and does not alone set has_issues.
recording_date_changed_recheck_required	Recording date changed; existing weather window must be rechecked.
unexpected_time_interval, unexpected_time_window, unexpected_row_count	Unexpected hourly window or row count; legitimate DST transitions are accepted.
duplicate_datetime	Unexpected duplicate instant, not the legitimate repeated DST hour.
missing_required_column, missing_datetime_column, unparseable_datetime, invalid_metadata_datetime	Missing or unreliable structure/time information; review.
empty_file, read_error; implausible_value	File unreadable/empty or a meteorological value is implausible.
Sentinel: missing_file, score_without_tif	Image missing, even if quality score exists.
too_cloudy_sentinel_not_available	Quality score equals 0; image too cloudy for use.
quality_score_missing	Quality assessment missing.
empty_file, raster_read_failed	Unreadable or empty Sentinel image.
inventory_not_run, qc_not_run	Inventory/QC missing, not equivalent to passing.